Executive Summary

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PART I – CREDIT SCORING

The analysis began with the evaluation of four financial indicators extracted from the historical sample of firms: Return on Sales (ROS), Leverage, Interest Coverage Ratio (ICR), and Customer Retention Rate (CRR). A univariate comparison between healthy and defaulted firms showed that CRR exhibited limited discriminatory power due to the strong overlap between the two groups. Consequently, the final discriminant model retained ROS, Leverage, and ICR as the core predictors of creditworthiness.

A Linear Discriminant Analysis (LDA) framework was then implemented to estimate the discriminant weight vector γ using the weighted variance-covariance matrix of healthy and distressed firms. The resulting coefficients were:

- ROS: **5.872**
- Leverage: **−1.911**
- ICR: **0.524**

These coefficients were used to compute a Z-score for each company, representing the firm's overall credit quality. The classification threshold α was defined as the midpoint between the average Z-scores of healthy and distressed firms:

- **α = 4.538**

The model generated the following classification errors:

- **Type I Error:** 3 distressed firms classified as healthy (3/10)
- **Type II Error:** 6 healthy firms classified as distressed (6/22)

Based on the empirical distribution of Z-scores, a four-category internal rating system was constructed:

- **A:** $Z > 6.5$
- **B:** $5 < Z \leq 6.5$
- **C:** $4 < Z \leq 5$
- **Defaulted:** $Z \leq 4$

Applying the model to the two loan applicants produced the following results:

- **ABC:** Z = 12.027 → Internal Rating: **A**
- **XYZ:** Z = 4.016 → Internal Rating: **C**

Given the bank's capital constraints and the significantly stronger financial profile of ABC, the loan was approved exclusively for ABC.

PART II – RATINGS AND CREDIT SPREAD

To estimate the probability of default and determine an appropriate loan yield for ABC, three complementary methodologies were implemented.

1. Rating-Based Approach (S&P Global)

Using S&P Global's historical cumulative default statistics for A-rated firms:

- **1-Year PD:** 0.05%
- **2-Year PD:** 0.11%

These values indicate a very low historical default probability consistent with investment-grade credit quality.

2. Merton (1973) Structural Credit Risk Model

The Merton structural framework was used to estimate the 2-year risk-neutral probability of default by modeling default as the event in which the firm's asset value falls below the face value of debt at maturity.

The following market inputs were used for ABC:

- Equity Market Value: **USD 2.747M**
- Face Value of Debt: **USD 3.700M**
- Asset Value: **USD 6.522M**
- Asset Volatility: **22%**
- Risk-Free Rate: **4%**
- Time Horizon: **2 years**

The model produced:

- **d1 = 2.235**
- **d2 = 1.924**
- **Risk-Neutral PD = 2.72%**

Compared to the historical rating-based approach, the Merton model generated a higher probability of default, reflecting its sensitivity to market volatility and capital structure assumptions.

3. Market-Based Comparable Bond Approach

Because ABC does not have publicly traded bonds, Starbucks Corporation was selected as a comparable issuer due to similarities in industry exposure, maturity profile, and investment-grade credit quality (BBB+).

Using the observed spread of the comparable bond:

- **Credit Spread:** 0.662%
- **2-Year U.S. Treasury Rate:** 4.00%
- **Implied Loan Yield:** 4.662%

Applied to a USD 100 million 2-year bullet loan, this corresponds to:

- **Repayment at Maturity:** USD 109.52M

Comparative Assessment of the Methodologies

The three methodologies provide complementary perspectives on credit risk estimation:

Approach	Strengths	Limitations
Rating-Based Approach	Simple, transparent, historically robust	Backward-looking and not firm-specific
Merton Model	Forward-looking and theoretically grounded	Highly sensitive to volatility and model assumptions
Market-Based Comparable Approach	Reflects current market conditions and investor sentiment	Dependent on peer selection and market liquidity

Conclusion

Across all methodologies, ABC consistently emerged as the financially stronger borrower and the most appropriate candidate for loan approval. The combined evidence from discriminant analysis, structural credit risk modeling, and market-based pricing supports the assignment of an internal **A-rating** to ABC.

Based on the comparable bond approach, the recommended pricing for the 2-year loan corresponds to an implied yield of **4.662%**.